

Real-time transcription of open language (ist)

Call the process

Obtain the authentication code: apply for appid from the iFLYTEK open platform, and add webapi dictation (streaming interface) to obtain the interface key APIKey and APISecret. Integrate the web socket interface: general interface + parameter description. Refer to the integration demo for implementation.

Competency Profile

It is a kind of speech recognition capability, which is used for real-time transcription technology for a long time, and supports real-time return of recognition results, so as to achieve the effect of text recognition while uploading audio.

Request address

```
ws[s]://ist-api.xfyun.cn/v2/ist?authorization=aG1hYyB1c2VybmFtZT0iMTAwSU1FIiwgYWxnb3Jp
dGhtPSJobWFjLXNoYTI1NiIsIGh1YWR1cnM9Imhvc3QgZGF0ZSB5ZXF1ZXN0LWxpbnUiLCBzaWduYXR1cmU9I1
VSbnk4M3o1e1JsNWF1OD11YXhUL1dGdUtWejZVNkdDdDV25SMGdueWc9Ig%3D%3D&date=Tue%2C+18+Dec+
2018+09%3A08%3A49+UTC&host=ist-api.xfyun.cn
```

Request Parameters and Meaning:

- Host: requested host, ist-API. Xfyun. CN.
- Date: current date, UTC time zone
- authorization: For authentication parameters, refer to [Shake hands](#)Right demo [demo-golang](#) [demo-java](#)

Data uploading and receiving

After the handshake is successful, the client and the server will establish a websocket connection, and the client can upload and receive data at the same time through the websocket connection. When the server has an identification result, it will push the identification result to the client through the websocket connection.

End marker:

The last frame of the audio needs to be marked with the end, status = 2:

```
{

  "data":{

    ..... # Other fields

    "status":2
```

```
}  
  
}
```

If no data is sent for more than 10 seconds, the server will actively disconnect the connection.

If the connection is closed for any reason in the session, the session is ended and unrecoverable. If the client continues to use the transcription service, it needs to re-establish the connection and open a new session. The server will use the status field as the last frame data identifier. When the status = 2, it indicates that all data has been sent. The client should stop sending data and disconnect immediately after processing the frame data. Details are as follows:

```
{  
  
  "data":{  
  
    "result":{..},  
  
    "status":2  
  
  }  
  
}
```

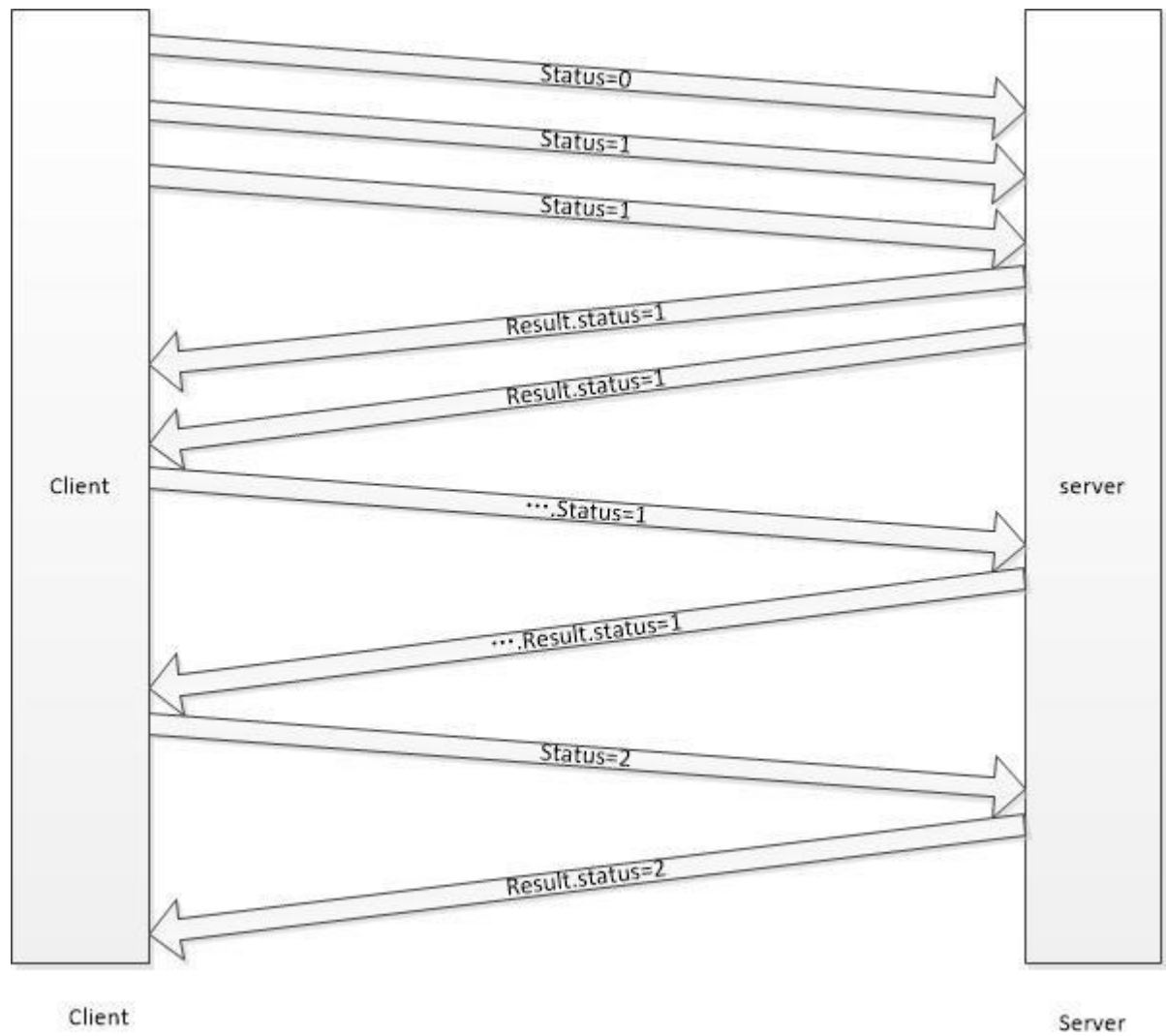


Figure 2.6.1 Data Interaction Process between Client and Server

An example of a request parameter

```
{
  "common":{
    "app_id":"123456"
  },
  "business":{
    "language": "zh_cn",
    "domain":"ist_ed_open",
```

```

    "accent":"mandarin"

  },

  "data":{

    "status":0,

    "audio":"exSI6ICJlbiIsCgkgICAgInBvc2l0aw9uIjogImZhbnHNlIgoJf..."

  }

}

```

Request parameter description

Description of common parameter: common mainly sets appid, uid and other parameters.
For more parameters, see [common](#)

Description of business parameters:

Parameter name	Type	Mu st pa ss on	Description
request_id	string	No	Mark the unique ID requested by the client. The maximum length is 64 characters. The client guarantees the uniqueness. This field is of great significance to the location request. It is recommended to pass it.
language	string	Yes	Input Chinese or English value range (zh_cn, en_us) Together with the domain, accent, and rate parameters, determine the transcription type. See the following for the specific relationship correspondence Rewrite type mappingTable
domain	string	Yes	Value range of domain (ist _ open, etc., refer to the list of languages)
language_type	int	No	Language filter filter. For ed engine 1: Chinese and English mode, both Chinese and English can be recognized 2: Chinese mode, simple English can be recognized 3: English mode, only English is recognized 4: Pure Chinese mode, only Chinese is recognized

			Default 1 Chinese and English mode
accent	string	Yes	Accent range (Mandarin, Cantonese)
dwa	string	No	Engine extension parameter, wpgs: enable the streaming result return function, limited function, unauthorized AppId cannot be used
eos	int	No	Used to set the silence time for endpoint detection in milliseconds
pd	string	No	Domain personalization parameters: law court Finance medical technology tech Sport education edu operator ISP Special customization sp _ care
res_id	string	No	Resource ID
vto	int	No	VAD forced-cut control in ms
punc	int	No	Punctuation control, default with punctuation, pass punc = 0 to close punctuation
nunum	int	No	Number warping is performed by default, and number warping is not performed when nunum = 0
pptaw	int	No	Segment control, set the number of words to be cached and then send to segment prediction
vgap	int	No	Unit audio frame, integer, range 0 to 180, default is 40 frames, set the clause interval, according to the business scenario settings, the recommended value is 20 to 40, not too large
personalizati on	object	No	PERSONAL (normal user resource); WFST (user patch big packet); LM (user patch small packet) The new personalized parameter supports transferring multiple resource ids at the same time in the form of key and value, such as "personalization": { "PERSONAL": "resid1", "L M": "resid2" }
seg_max	int	No	The post-processing segmentation function controls the maximum number of words (0-500)
seg_min	int	No	The post-processing segmentation function controls the minimum number of words (0-50)
seg_weight	float	No	The post-processing segment function controls the segment length weight (0-0.05)
speex_size	int	No	Frame rate for speex audio, used only for speex audio

			1 When the speex code is not customized by iFLYTEK, it must be specified. 2 When speex code is customized for iFLYTEK, do not set it. Note: Please refer to here for the customized SPEEX coding tool of iFLYTEK. Speex encoding.
spkdia	int	No	Real-time transliteration of role separation mode. 0: Turn off speaker separation 1: Turn on real-time turning point 2: Turn on real-time character separation.

List of languages

Access parameter: language accent domain sample _ rate is a four-tuple key value, and the value value corresponding to each dialect is as follows

Audio sampling rate: 16K audio

Permission request: Please go to itesm: <https://itsm.iflytek.com/> (AI Cloud Service-Permission Application) Submit the requisition application

language	accent	domain	Type of transcription
en_us	mandarin	ist_open	English
zh_cn	mandarin	ist_ed_open	A mixture of Chinese and English
zh_cn	cantonese	ist_open	Cantonese
ru_ru	mandarin	ist_open	Russian
es_es	mandarin	ist_open	Spain
ja_jp	mandarin	ist_open	Japanese
fr_fr	mandarin	ist_open	French
ar_il	mandarin	ist_open	Arabic
vi_vn	mandarin	ist_open	Vietnamese
ko_kr	mandarin	ist_open	Korean
zh_cn	mandarin	ist_ed_open	Chinese-English mixed quotation

			Engine, transitional use
it_IT	mandarin	ist_open	Italian
th_TH	mandarin	ist_open	Thai

Parameter name	Type	Must pass on	Description
status	int	Yes	Used to distinguish data States, 0 for first block of audio, 1 for middle block of audio, and 2 for last block of audio
format	string	Yes	The sampling rate of audio supports 16K and 8K, 8K will be processed by 16K, 16K audio: audio/L16; rate = 16000, 8K audio: audio/L16; rate=8000
encoding	string	Yes	Audio data format, raw identifies native audio, and speex represents audio compressed by speex. Supported compression formats also include: opus-ogg, opus-wb
audio	string	Yes	Audio content, base64 encoded

An example of a return parameter

Default format (wpgs on)

```
{
  "code": 0,
  "message": "success",
  "sid": "ist00dad3b@dx179b29704936f22822",
  "data": {
    "status": 1,
    "result": {
      "sn": 4,
      "ls": false,
      "bg": 12620,
```

```
"ed": 13450,

"pgs": "apd",

"sub_end": true,

"rg": [1,3],

"ws": [{

    "bg": 1,

    "cw": [{

        "rl": 0,

        "sc": 0.00,

        "W": "a while",

        "wb": 1,

        "wc": 0.00,

        "we": 36,

        "wp": "n"

    }]

}, {

    "bg": 37,

    "cw": [{

        "rl": 0,

        "sc": 0.00,

        "W": "just",

        "wb": 37,

        "wc": 0.00,

        "we": 52,
```



```
        "wp": "n"

    }]

}, {

    "bg": 53,

    "cw": [{

        "r1": 0,

        "sc": 0.00,

        "W": "Stop",

        "wb": 53,

        "wc": 0.00,

        "we": 64,

        "wp": "n"

    }]

}, {

    "bg": 65,

    "cw": [{

        "r1": 0,

        "sc": 0.00,

        "W": "yes",

        "wb": 65,

        "wc": 0.00,

        "we": 72,

        "wp": "n"

    }]

}
```

```

    }}

  }

},

"context_id": "aGZpc3SSHwoBVzWGKYwqf2ReFgUA"

}

```

Definition of JSON field:

"" Sn "": 4, //serial number of recognition result

"" Ls "": false, //is the last frame result

"" Pgs "": "" APD "" //Is the currently recognized sentence a replacement or an append? APD means append and RPL means replace

"rg": [1, 3], //replacement scope

"bg": 29332, //start time of the sentence

"ed": 34523 //End time of the sentence

"ws": [//sentence

```

    {

```

"bg": 278, //sentence start position, position is
relative to sentence BG

"cw": //word recognition result

```

        [{

```

"sc": 0.0, //sentence confidence

"w": "an",

"wb": 278, //Number of frames the word
starts (note 10 ms per frame), position is relative to BG in the VAD result

"wc": 0.0, //the confidence level of the word

"we": 299, //The number of frames the word ends in,
relative to the VAD knot

The ed in fruit

```

"wp":"n" //attribute of a word, where n represents a normal word, s,
Indicates smoothness, p indicates punctuation, and
G indicates segmentation

    }}

    },

]

```

Return parameter description

Parameter	Type	Description
sid	string	Unique identification for each session
code	int	Error code, 0 identification success
context_id	string	Business context ID, which uniquely identifies a complete conversation (including successful reconnection requests)
message	string	Error description
data	object	Identification data
data.status	int	Identification of whether the identification result is finished: 1: identifying 2: End of recognition
data.result	object	Recognition results
data.result.sn	int	The sequence number of the returned result
data.result.pgs	string	When wpgs is enabled, there will be this field. When the value is "" APD "", it means that this piece of result is appended to the previous final result. When the value is "" RPL "", it means that some of the previous results are replaced. The replacement range is the rg field
data.result.rg	array	Replace the scope, turn on wpgs to have this field,

data.result.ls	bool	Is it the last result?
data.result.bg	int	Time unit of the beginning of the audio segment, ms
data.result.ed	int	Time unit of the end of the audio segment ms
data.result.sub_end	bool	Secondary reserved field that identifies whether the current clause has ended and does not need to be parsed by the client
data.result.ws	array	Sentence
data.result.ws.bg	int	Sentence start position, relative to Data. Result. BG , unit frame, 10ms per frame
data.result.ws.cw	array	Sentence recognition result
data.result.ws.cw.rl	uint64	Current speaker role ID. When rl is non-zero after the role separation function is enabled, it indicates that the current speaker role has been switched.
data.result.ws.cw.sc	float	Sentence confidence
data.result.ws.cw.w	string	The content of words
data.result.ws.cw.wb	int	Number of frames at the beginning of a word, 10ms per frame
data.result.ws.cw.wc	float	Confidence of words
data.result.ws.cw.we	int	The number of frames the word ends in
data.result.ws.cw.wp	string	Attribute of a word, where n represents a normal word, s represents smoothness, p represents punctuation, and G represents segmentation

If an error is encountered, refer to the list of error codes, as shown in [List of error codes](#)

Common error codes

- 10163: Parameter verification failed. This is caused by the failure of client parameter verification. The client needs to change the request parameters according to the description in the returned message field
- 10313: The first frame of the request parameter does not transmit the app _ ID or the transmitted app _ ID does not match the API _ key.
- 10043: The audio decoding failed. Please check whether the transferred audio corresponds to the encoding format described in the encoding field.
- 11201: The interface usage exceeds the maximum limit of purchase. Please continue to use it after purchase. Internal users can submit a demand work order (itsm. Iflytek. Com -- >

AI Cloud Service -- > 1001-AI Capability Usage Application) on itsm to apply for additional usage.

- 10114: The request timed out and the session time exceeded 60000s. Please control the session time and keep it below 60000s

Frequently asked questions

- The server returns the recognition result, but it is too different from the actual result of the input audio. Solution: 1. The server only supports audio with a sampling rate of 16K and 8K. Check whether the sampling rate of the audio meets the requirements, and check whether the sampling rate in the format parameter is consistent with the actual sampling rate of the audio. Audio only needs to be uploaded after base64 encoding. If you use scripting languages such as JS, make sure that the audio is only base64 encoded. 3 If the audio is in a compressed format, ensure that the compression format is consistent with the encoding parameter.
- Audio coding can refer to DEMO:[demo](#)

Refer to when you encounter other errors [List of error codes](#)